2024

Trust Bank Builds a Scalable and Innovative Digital Bank on AWS

Learn how Trust Bank built a scalable digital bank while creating an innovative customer experience using the AWS Cloud.

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3-minute

onboarding experience

87% lower cost

of acqu

10x increase

in maximum number of database I/O c

67% reduction

in maximum P99 latency for payment processing

Overview

When Trust Bank launched in 2022, Singapore was arguably oversaturated with banking options, with 98 percent of the population having at least one bank account. Yet, Trust Bank reached more than 100,000 customers within 10 days of its launch. In its first year, it onboarded around 14 percent of the market, making it one of the world's fastest growing digital banks.

New user setup was setting benchmarks, taking less than 3 minutes and with a straight-through processing rate higher than 90 percent. Trust also rolled out a broad range of products from the start, including current deposit accounts, credit and debit cards, and an extensive rewards platform. Few banks achieve such a well-developed customer proposition this quickly. To support this rapid growth, the bank was built entirely on Amazon Web Services (AWS).

Using Amazon EKS to Unlock Resilience and Scalability

To deliver a compelling customer experience at scale, Trust Bank needed agile, reliable, and secure infrastructure. It chose Amazon Elastic Kubernetes Service (Amazon EKS), a managed Kubernetes service, to run its services.

The bank runs nine Amazon EKS clusters and deploys all workloads across three Availability Zones to improve resilience. Moreover, it deploys applications with static stability so that it has enough capacity if a node or Availability Zone becomes unavailable. Amazon EKS cluster upgrades are managed using a blue/green deployment strategy to minimize downtime.

Meanwhile, core banking components from Thought Machine, an AWS Partner, are automatically scaled from 400 to 600 pods during daily peak hours. The bank scales cluster nodes 10 times faster using Karpenter, an open-source high-performance Kubernetes cluster automatic scaler.

Migrating to Amazon Aurora for PostgreSQL to Increase I/O and Reduce Latency

Six months after its launch, Trust Bank decided to improve resiliency and increase input/outpu[...]nce by migrating from Amazon RDS for PostgreSQL—which gives companies access to the familia database engine—to Amazon Aurora, which provides unparalleled high-performance and availability at global scale with full MySQL and PostgreSQL compatibility, at 1/10th the cost of commercial databases.

By adopting AWS Graviton processors, which provide up to 40 percent better price performance compared with the corresponding x86-based instances, Trust Bank reduced database costs by 18 percent, even with 30 percent more customers. It also reduced the I/O cost on Amazon Aurora and increased the maximum I/O operations per second by 10 times. Now, P99 latency for payment processing is 67 percent lower.

Using Machine Learning to Make Intelligent Recommendations for Customers

Trust Bank's data lake stores hundreds of terabytes of data on Amazon Simple Storage Service (Amazon S3), an object storage built to store and retrieve any amount of data from anywhere. Data lake inputs stream in near real time from Apache Kafka—an open-source distributed event streaming system—to Amazon S3 using Apache Kafka Connect. Data is then used for business intelligence and machine learning (ML) use cases.

Trust Bank implemented a data mesh model to manage data governance and data permissions. In addition, all sensitive data are tokenized in the data lake.

Trust Bank is deploying ML to enhance operations and provide personalized customer recommendations using Amazon SageMaker, a fully managed ML service. It's also using Amazon EMR, a cloud big data solution, for training and inference. "We're innovating and releasing new products much faster due to our life cycle on AWS," says Rajay Rai, chief information officer at Trust Bank.

About Trust

Digitally native Trust Bank was founded in Singapore in 2022. Backed by Standard Chartered Bank and FairPrice Group, it became one of the world's fastest-growing digital banks by market share in its first year.

AWS Services Used

Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

Learn more »

Amazon Aurora

Amazon Aurora is a relational database service that combines the speed and availability of high-end commercial databases with the simplicity and cost-effectiveness of open-source databases. Aurora is fully compatible with MySQL and PostgreSQL, allowing existing applications and tools to run without requiring modification.

Learn more »

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to enable high-performance, low-cost machine learning (ML) for any use case.

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Amazon EMR

Amazon EMR is the industry-leading cloud big data solution for petabyte-scale data processing, interactive analytics, and machine learning using open-source frameworks such as Apache Spark, Apache Hive, and Presto.

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